Activation Energy and Catalysts

Answer key with worked solutions

A. Understanding

1. Definitions.

> **Activation energy**: the minimum energy that colliding particles need in order to react. **Transition state**: the highest-energy, unstable arrangement of atoms on the energy profile, between reactants and products.

2. Energy profile.

> (a) ΔH is negative, so **exothermic**. (b) Ea (reverse) = 120 kJ/mol + 40 kJ/mol = **160 kJ/mol**. (c) The catalyst lowers both hills by 50 kJ/mol: Ea (reverse) = 70 kJ/mol + 40 kJ/mol = **110 kJ/mol**; ΔH is **unchanged, −40 kJ/mol**.

3. Maxwell–Boltzmann.

> Only molecules with energy ≥ Ea can react. At a higher temperature the distribution flattens and shifts to higher energy, so the **area beyond Ea grows much more than the average energy does**. A small temperature rise can therefore roughly double the number of molecules able to react (and they collide slightly more often too).

4. Catalysts.

> A **homogeneous** catalyst is in the same phase as the reactants (e.g. H^+ ions catalysing ester hydrolysis in solution). A **heterogeneous** catalyst is in a different phase, usually a solid surface (e.g. iron in the Haber process, platinum in catalytic converters). A catalyst **does not change K**.

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B. The Arrhenius equation

5. Factor from 300. K to 310. K.

$$\ln\frac{k_2}{k_1} = \frac{7.50\times10^{4}\ \text{J/mol}}{8.314\ \text{J/(mol·K)}}\left(\frac{1}{300.\ \text{K}} - \frac{1}{310.\ \text{K}}\right) = 0.970$$

$\frac{k_2}{k_1} = e^{0.970} = \mathbf{2.64}$ (the units cancel inside the exponent)

6. Ea from two rate constants.

$$\ln\frac{6.0\times10^{-4}\ \text{s}^{-1}}{3.0\times10^{-5}\ \text{s}^{-1}} = \ln 20. = 2.996;\ \frac{1}{500.\ \text{K}} - \frac{1}{550.\ \text{K}} = 1.818\times10^{-4}\ \text{K}^{-1}$$

$$E_\text{a} = \frac{8.314\ \text{J/(mol·K)}\times 2.996}{1.818\times10^{-4}\ \text{K}^{-1}} = 1.37\times10^{5}\ \text{J/mol} = 137\ \text{kJ/mol}$$

The rate constants have 2 significant figures, so report **Ea = 1.4 × 10² kJ/mol** (137 kJ/mol before rounding).

7. k at 330. K.

$$\ln\frac{k_2}{k_1} = \frac{7.50\times10^{4}\ \text{J/mol}}{8.314\ \text{J/(mol·K)}}\left(\frac{1}{300.\ \text{K}} - \frac{1}{330.\ \text{K}}\right) = 2.734$$

$k_2 = 2.0\times10^{-4}\ \text{s}^{-1}\times e^{2.734} = 3.1\times10^{-3}\ \mathbf{s^{-1}}$ ($3.08\times10^{-3}\ \text{s}^{-1}$ before rounding to 2 significant figures)

C. Challenge

8. Effect of the catalyst.

With A unchanged, $\frac{k_\text{cat}}{k} = e^{(E_\text{a}-E_\text{a,cat})/RT}$:

$$\frac{(7.50-5.00)\times10^{4}\ \text{J/mol}}{8.314\ \text{J/(mol·K)}\times 298\ \text{K}} = 10.09$$

$\frac{k_\text{cat}}{k} = e^{10.09} = 2.4\times10^{4}$: about 24 000 times faster.

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9. Arrhenius plot.

> Slope $= -\dfrac{E_a}{R}$, so $E_a = 6.65 \times 10^{3}\ \text{K} \times 8.314\ \text{J/(mol·K)} = 5.53 \times 10^{4}\ \text{J/mol} = \mathbf{55.3\ kJ/mol}$ (the units of K cancel).

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